

Point-star astrometry across a fisheye field

A Barghini-model feasibility result

Peter Thejll • 13 September 2026

An untrailed 1569×1444 -pixel Milky Way image yields 3,688 measured sources and 3,653 fitted catalogue associations. The root-mean-square (RMS) positional residual is 0.398 px, with median 0.249 px. Figure 1 shows the agreement across the observed sky; Figure 2 resolves the remaining discrepancies.

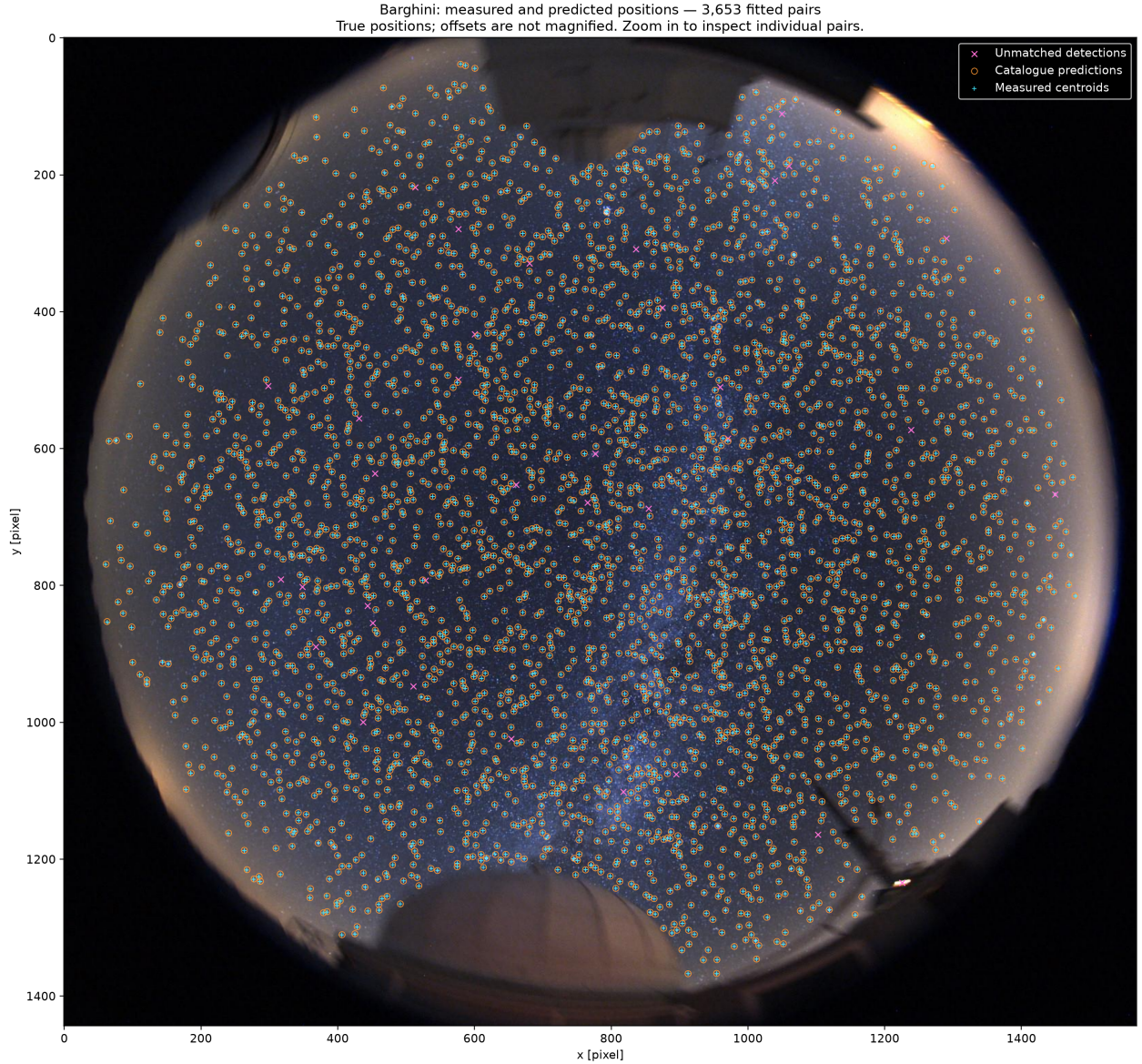


Figure 1: Measured centroids (cyan crosses) and catalogue positions projected through the fitted Barghini model (orange circles), for all 3,653 fitted pairs. Pink crosses mark 35 unmatched detections. Coordinates and separations are displayed at their true image positions; no cosmetic shifts or residual magnification are applied. The full-resolution PDF supports inspection by zooming.

Centre-to-edge behaviour

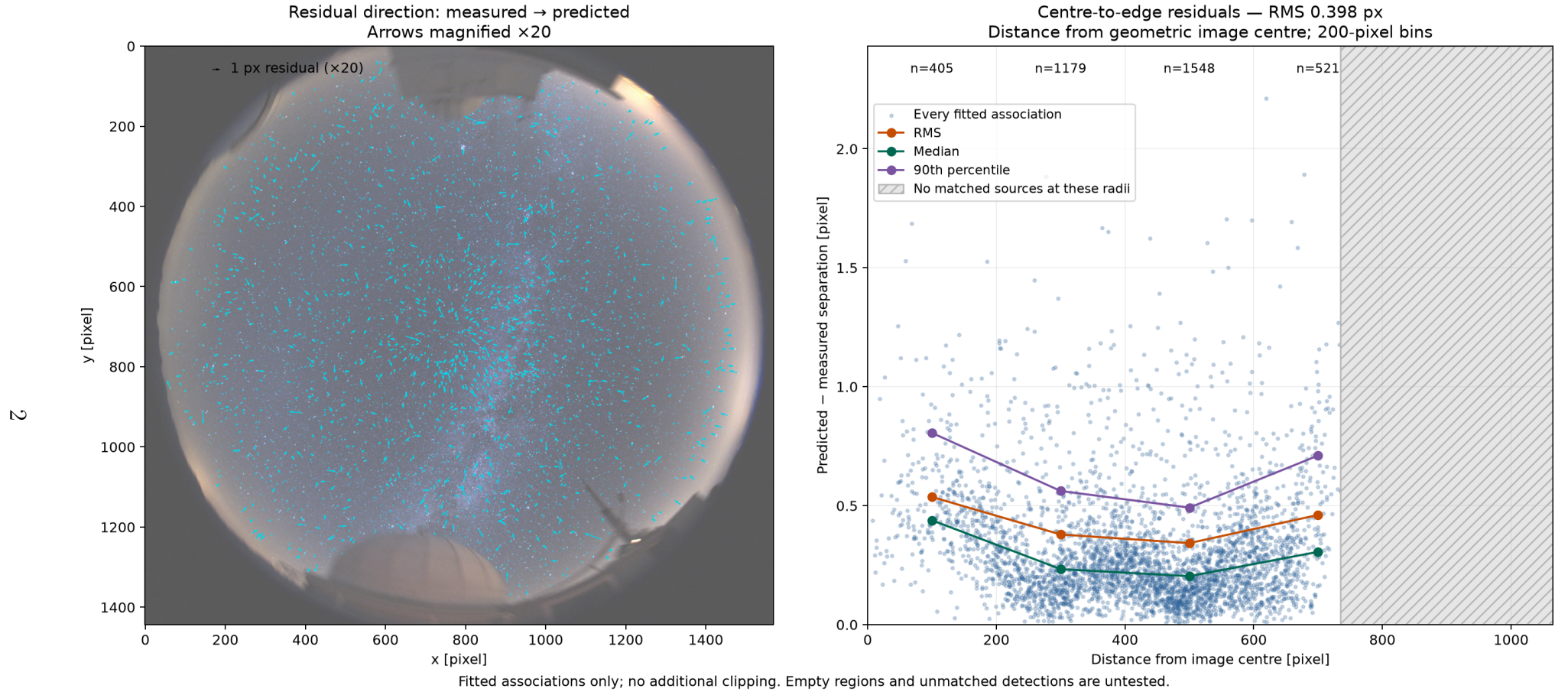


Figure 2: Left: vectors from measured to predicted positions, magnified $\times 20$ solely for visibility. Right: separation for every fitted association, with RMS, median and 90th percentile in 200-pixel annuli about the geometric image centre. The hatched region has no matched sources. No additional residual clipping is applied. These diagnostics describe fitted associations, not independent validation.

The outer band shows a modest RMS rise, without progressive edge divergence among the matched sources. Small coherent residual patterns remain visible. Any future correction must be made through the astrometric model and its exported coordinates; plotting must preserve the resulting discrepancies.

Method and quantitative result

Centroids are measured directly from the image after local background subtraction. Growing measurement windows retain broad and saturated sources. A blind `tetra3` pattern match supplies 11 initial star associations from a small patch. Progressive matching against the frozen Tycho-2 catalogue, supplemented with bright Hipparcos stars, then expands the fitted field. The global projection uses the independent O/Z parametrisation of Barghini et al. (2019), equations (5), (6) and (11), with the inherited fish-eye transform discussed on page 4:

$$u(r_O) = Vr_O + S[\exp(Dr_O) - 1],$$

where r_O is distance from the optical centre. The diagnostic radius in Table 1 is instead measured from the geometric image centre, (784.0, 721.5) pixels.

Table 1: Residuals for all fitted associations in each image-centred annulus. Radial bias is the mean signed component of predicted minus measured position, positive outwards. All residual columns are in pixels.

Radius [px]	Stars	RMS	Median	90th pct.	Radial bias
0–200	405	0.536	0.439	0.806	+0.329
200–400	1,179	0.379	0.233	0.562	−0.068
400–600	1,548	0.343	0.204	0.492	+0.015
600–800	521	0.461	0.306	0.711	−0.040

Interpretation. The largest fitted separation is 2.211 px; the outermost matched source is at radius 734.75 px, so the last populated annulus is only partly sampled. The inner-band outward bias of 0.329 px shows that systematic structure remains. All sources were available for fitting, with no stars withheld. Associations were selected with a three-pixel gate before the final fit. Unmatched sources and unsampled regions have no evaluated star-pair residual. The results therefore establish the feasibility of this full-field fit, with conditional fit residuals rather than an independent accuracy or completeness estimate. Observation-epoch propagation, atmospheric refraction and terrestrial orientation are outside this experiment.

Comparison and next steps. In the recorded, time-bounded comparison, direct whole-image `solve-field` attempts were unsolved. On a 260×260 -pixel crop, the same measured dots gave 0.390 px RMS with third-order SIP distortion, against Barghini’s 0.406 px on the same 195 stars. This supports a benefit in field coverage while demonstrating comparable local precision. Further work should test other images and lenses, and evaluate a bright-star supplement plus Gaia DR3/DR4 in place of Tycho, with epoch propagation.

Scientific lineage and scope of this implementation

Barghini et al. (2019) explicitly place their method within earlier fireball-camera astrometry in their Section 2. The principal development is:

Cep-lecha (1987). The empirical radial description $z = U + Vr + S \exp(Dr)$ accounts for the strong change in image scale towards the horizon. Barghini and colleagues credit Cep-lecha (1987) with this starting point for the all-sky model.

Borovička (1992). The subsequent formulation separates projection coordinates about the optical axis from horizontal celestial coordinates and gives the constrained radial law $u = Vr + S[\exp(Dr) - 1]$ (Borovička, 1992). This eight-parameter model is also reproduced explicitly in Section 1.2 of Borovička et al. (1995).

Borovička, Spurný and Keclíková (1995). Their tests on guided all-sky photographs exposed systematic near-horizon errors in the preceding model. They introduced an additional exponential term in squared radius and corrections for plate inclination (Borovička et al., 1995). Their Figures 2–3 demonstrate the same diagnostic principle used here: examine residuals across the field and correct the astrometric transformation when the evidence requires it.

Barghini and colleagues (2019). Their Sections 4.2–5 identify strong coupling between optical-centre coordinates and the angular offset to the zenith. Replacing the latter by explicit detector coordinates of the zenith yields the O/Z parameterisation used here. They also describe initialisation from meridian crossings, progressive source association and centroiding tests (Barghini et al., 2019). Thus the radial projection has a clear lineage, while the 2019 work improves its parameterisation and practical calibration.

What this experiment adopts. Our single-image solver combines the analytical O/Z model with its own dot finder, tetra3 bootstrap and robust fit. It does not reproduce the paper’s multi-epoch meridian-crossing initialisation: Z is a fixed celestial reference direction here, with no inferred local zenith. The extra 1995 correction terms and Barghini’s Section 6.4 tabulated systematic corrections are also absent. The remaining residuals are displayed faithfully; any added calibration must be part of the sky–pixel mapping and saved coordinates.

Reproducibility. The [private WIDE-FIELD-SOLVER repository](#) preserves the original solver at [v0.1.0](#). The diagnostics retain the same camera and coordinates. Both figures are included unchanged. The photograph’s original attribution and redistribution terms are undocumented; this report is for private research use.

References

- Dario Barghini, Daniele Gardiol, Albino Carbognani, and Salvatore Mancuso. Astrometric calibration for all-sky cameras revisited. *Astronomy & Astrophysics*, 626:A105, 2019. doi: 10.1051/0004-6361/201935580. URL <https://doi.org/10.1051/0004-6361/201935580>.
- Jiří Borovička. Astrometry with all-sky cameras. *Publications of the Astronomical Institute of the Czechoslovak Academy of Sciences*, 79:19–22, 1992. URL <https://www.astronom.cz/borovicka/publik.htm>. Bibliographic record in the author’s publication list.
- Jiří Borovička, Pavel Spurný, and J. Keclíková. A new positional astrometric method for all-sky cameras. *Astronomy and Astrophysics Supplement Series*, 112:173–178, 1995. URL <https://articles.adsabs.harvard.edu/pdf/1995A%26AS..112..173B>.
- Zdeněk Cep-lecha. Geometric, dynamic, orbital and photometric data on meteoroids from photographic fireball networks. *Bulletin of the Astronomical Institutes of Czechoslovakia*, 38:222–234, 1987. URL <https://ui.adsabs.harvard.edu/abs/1987BAICz..38..222C>.